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Brief Communication

Artificial intelligence-driven automated lung sizing from chest radiographs

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ABSTRACT

Lung size measurements play an important role in transplantation, as optimal donor-recipient size matching is necessary to ensure the best possible outcome. Although several strategies for size matching are currently used, all have limitations, and none has proven superior. In this pilot study, we leveraged deep learning and computer vision to develop an automated system for generating standardized lung size measurements using portable chest radiographs to improve accuracy, reduce variability, and streamline donor/recipient matching. We developed a 2-step framework involving lung mask extraction from chest radiographs followed by feature point detection to generate 6 distinct lung height and width measurements, which we validated against measurements reported by 2 radiologists (T.A. and W.B.G.) for 50 lung transplant recipients. Our system demonstrated <2.5% error (<7.0 mm) with robust interrater and intrarater agreement compared with an expert radiologist review. This is especially promising given that the radiographs used in this study were purposely chosen to include images with technical challenges such as consolidations, effusions, and patient rotation. Although validation in a larger cohort is necessary, this study highlights artificial intelligence's potential to both provide reproducible lung size

Abbreviations: AI, artificial intelligence; CXR, chest radiograph.

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assessment in real patients and enable studies on the effect of lung size matching on transplant outcomes in large data sets.

1. Introduction

Although lung transplantation is the only effective treatment for end-stage lung disease, precise donor/recipient lung size matching is essential for optimal therapeutic outcomes.[1-3] Several methods exist to optimize lung size matching—including 2-dimensional size measurement on chest radiographs (CXRs),[4-6] predicted total lung capacity (pTLC),[7,8] and volumetric assessments using computed tomography (CT) imaging.[3,9] Each has its limitations, however, and there is no consensus best method. Eg, pTLC can be misleading if height is reported inaccurately and does not account for weight, race, or disease—all of which affect chest cavity size. Volumetric assessments have not been easily implemented outside of research applications, and hemodynamic instability may preclude their acquisition. Finally, CXR measurements, although having been shown to correlate well with actual total lung capacity (TLC),[5,10] are prone to technique-related errors including rotation, effusion, or consolidation that obscures images and are also inevitably affected by reader subjectivity despite that fact that averaging multiple radiographs does decrease this error.[4] Given these challenges, there is a pressing need to develop automated approaches that both provide standardized lung size measurements and are capable of processing large data sets to compare different size matching strategies. A similar need for improved precision medicine approaches in other fields has prompted an increased focus on using artificial intelligence (AI) for computer-assisted lung assessment[11,12]: eg, for diagnosis and treatment planning. We therefore sought to develop a scalable, easy-to-use automated system capable of accurately reporting standardized metrics used for donor/recipient lung size matching.

2. Materials and methods

This study was conducted under an institutional review board approved protocol (#848398). A novel 2-step framework for obtaining automated lung size measurements was developed, as shown in Figure 1: (1) automatic lung segmentation on CXRs and (2) computer vision postprocessing to identify 6 distinct size measurements. In step 1, we extracted lung masks from posteroanterior CXRs using PSPNet,[13] a segmentation model capable of exploiting the image's global context information, which not only looks at the individual pixels around the lung boundary but also considers the overall structure and shape of the entire thoracic cavity, making it possible to accurately identify the lung boundaries even in the presence partial obstructions.

We trained the model on the publicly available Darwin data set,[14] comprising 6500 CXRs and lung masks. Lung segmentations were initially generated by Darwin autoannotate AI and subsequently reviewed and fine-tuned by human annotators. To enhance the model's robustness and generalizability across various scenarios, data augmentation techniques such as rotation, scaling, and translation were applied. Additionally, contrast-limited adaptive histogram equalization was used to improve image contrast, further aiding the segmentation process. In step 2, computer vision techniques were used to automatically identify crucial feature points from the extracted lung masks, enabling acquisition of the following measurements: right and left lung apex-to-costophrenic angle, right and left apex-to-diaphragm, as well as lung width at both hilum and diaphragm. These feature points were defined as follows:

1. Lung apex: a point with the highest y-coordinate of the lung contour and an x-coordinate equal to the average of the x-coordinates along the top 20% of that contour.
2. Lung midpoint: the average x- and y-coordinates of the lung contour, representing the midpoint inside the lung.
3. Costophrenic angle point: the farthest point from the lung midpoint on the lateral part of the lung contour. A point on the lateral part of the lung contour that has the same y-coordinates as the shorter of the costophrenic angles of the 2 lungs was used to ensure horizontal measurement accuracy.
4. Mid-diaphragm point: a point on the inferior part of the lung contour with the same x-coordinate as the lung midpoint.
5. Lateral midpoint: a point positioned on the lateral part of the lung contour sharing y-coordinates with the average of the midpoints of the 2 lungs (ensuring horizontal measurements).

Lung size measurements were generated using the distances between these defined feature points for reference, without considering height or weight, to ensure standardization across patients. We validated our automated measurements against those independently reported by 2 radiologists (T.A. and W.B.G.) in a blinded setting for 50 portable CXRs acquired in lung transplant recipients at the University of Pennsylvania (demographic information summarized in Table 1). As shown in

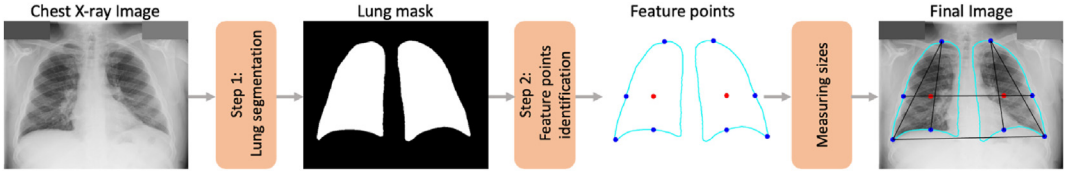

Figure 1. Schematic illustration of the proposed framework. First, lung mask segmentation is extracted from chest radiographs. Second, computer vision techniques are used to identify the lung apices, costophrenic angles, mid-diaphragm points, and lateral midpoints. The distances between these points, shown in the final image, are calculated and reported in millimeters.

Table 1
Cohort demographics (N = 50).

Demographic	Value
Age (y)	59.4 ± 10.8
Male	35 (70)
Race	
White	38 (78)
Black	3 (6)
Hispanic	3 (6)
Asian	4 (8)
Native American	1 (2)
Other	5 (10)
Height (inches)	67.8 ± 4.1
Weight (lb)	176.1 ± 33.7
BMI (kg/m^2)	26.9 ± 4.3
Image classification	
Normal	27 (54)
Abnormal	
Pleural Effusion	8 (16)
Atelectasis	10 (20)
Consolidation	1 (2)
Opacities	4 (8)

Continuous variables expressed as mean ± SD and categoric variables as n (%).
BMI, body mass index.

Figure 2, this data set included images displaying consolidation, infiltrates, and effusion, as well as technical issues such as rotation. Linear regression was then performed to further align the calculated measurements with the average of the radiologists' measurements. We further calculated the *r* value of linear regression to demonstrate agreement between AI-generated and radiologist measurements; the intrarater correlation coefficient was calculated to assess measurement consistency between radiologists.

3. Results

Automated standardized lung segmentations and measurements were successfully obtained for all subject images. The average error between automated measurements and the average of 2 separate radiologist-generated measurements was less than 2.5% (<7.0 mm) (Fig. 3), and interrater agreement between AI and radiologists was almost identical to that observed between radiologists alone, with a Pearson correlation coefficient of 0.97. Intrarater correlations, both between radiologists (T.A. and W.B.G) and between radiologists and AI, also exceeded 0.90, underscoring the consistency and reliability of our automated measurements (Table 2).

4. Discussion

This pilot study demonstrated that deep learning and computer vision are powerful tools for generating standardized lung measurements in a cohort of recipients with both normal and abnormal images (including effusion, consolidation, and patient rotation). Performing comparably with expert radiologists' readings from CXRs, this approach not only provides accurate and consistent measurements, but also has the potential to dramatically streamline clinical workflows, reduce the burden on radiologists, and improve overall patient care. Our system's ability to rapidly perform all lung measurements in seconds, without the need for radiologist availability or technical knowledge, has significant potential to enhance clinical efficiency and decrease preprocurement evaluation time. This is particularly relevant given that the time from electronic donor offer to cross clamp has

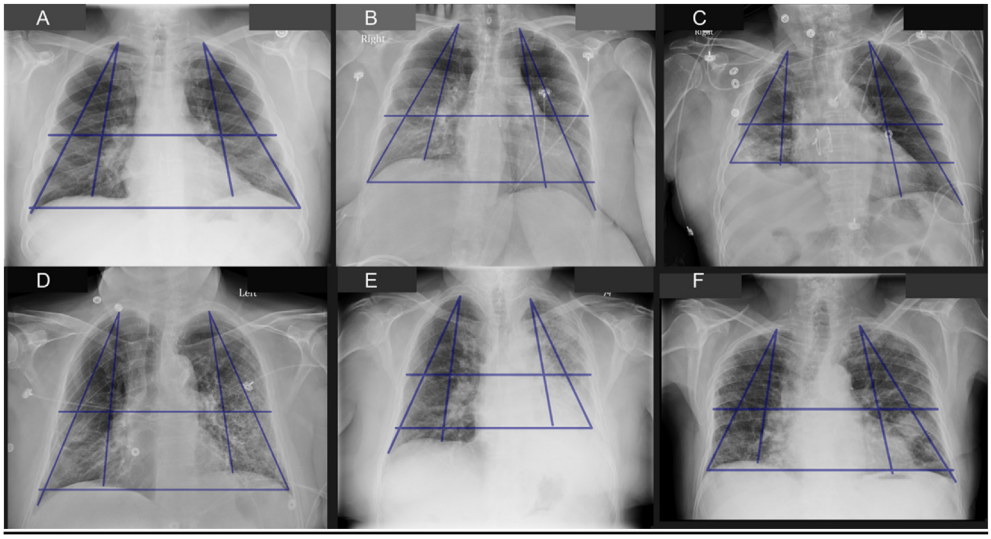

Figure 2. Representative chest radiographs with automated measurements. Our data set includes chest radiographs without lung abnormalities (A,B) as well as with unilateral or bilateral lung opacities (C-F), pleural effusions (C,E), and technical issues such as rotation (C,D). Despite the presence of these conditions, the automated measurements were executed successfully, exhibiting no discernible errors.

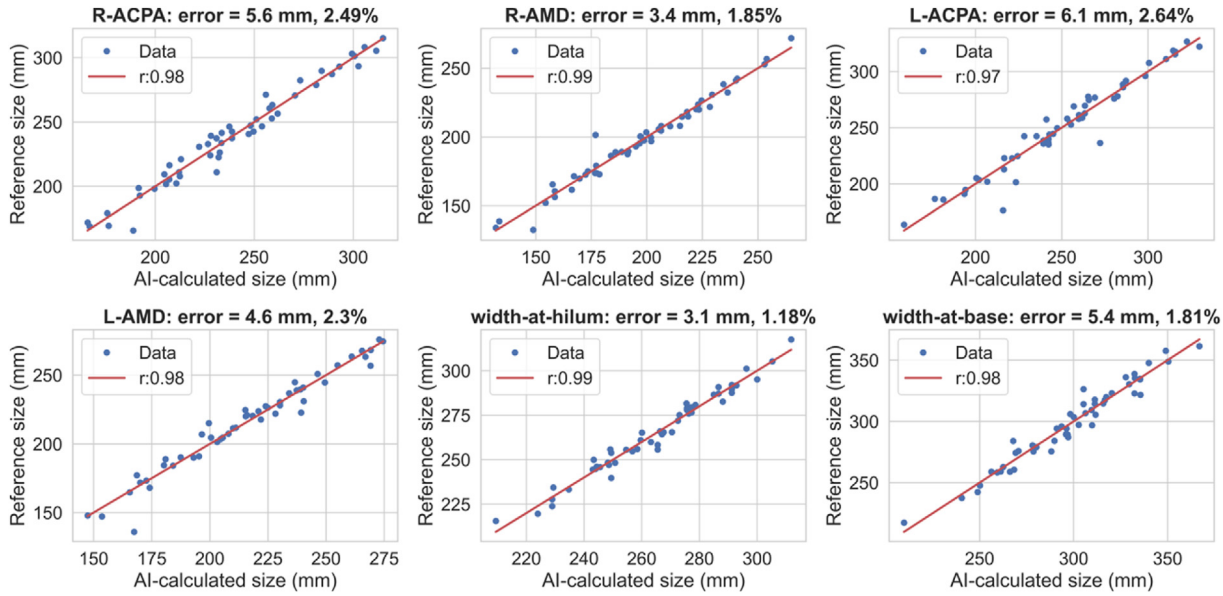

Figure 3. Comparison of reference size (average of 2 radiologists) vs artificial intelligence measurements, showing an average error of less than 2.5% (<7.0 mm). L-ACPA, left lung apex-to-costophrenic angle; L-AMD, left apex-to-diaphragm; R-APCA, right lung apex-to-costophrenic angle; R-AMD, right apex-to-diaphragm.

increased from 29 to 32 hours with initiation of continuous distribution[15]—an increase that also highlights the difficulties in assessing,[4] managing, and obtaining suitable organs for transplant. Our rapid, standardized assessment system also opens new avenues for research on donor/recipient size-mismatch and size-matching strategies using large imaging data sets.

Despite the excellent reliability and consistency demonstrated between our automated measurements and those of radiologists, there were several particularly challenging cases—such as in the presence of pleural effusion, which can obscure the costophrenic angle—when both radiologists encountered difficulties in accurately assessing lung size and had to use their best judgment. As a result, these correlations cannot necessarily be taken to indicate correct measurements. Further validation in a larger cohort is necessary to establish the system's applicability across diverse clinical scenarios and to address potential influences from lung pathologies, as well as technical issues in portable CXR acquisition including rotation, distance from the source, and cutoffs at the apex or sides. It must also be acknowledged that, despite its ubiquity, CXR interpretation presents notable challenges, including interpretational variability, technical limitations, and the potential for human error; indeed, overall error rates are estimated to be between 4% and 30% dependent on the underlying prevalence of lung abnormalities in the population studied.[16] With respect to lung measurements in particular, the reported error rate has varied between 4.0 and 8.0 cm depending on the specific measurement.[4] However, because radiologist assessment remains the gold standard for interpretation, it represents the reference point against which our automated system's performance must be judged.

Given that our automated tool does not currently report any technical challenges it encounters, transplant clinicians should

Table 2
Intraclass correlation coefficient with CI as a measure of raters' reliability.

Comparators	R-ACPA	R-AMD	L-ACPA	L-AMD	Width-at-hilum	Width-at-base
Radiologist 1 (T.A.) vs radiologist 2 (W.B.G.)	0.98 (0.96, 0.99)	0.96 (0.93, 0.97)	0.96 (0.94, 0.98)	0.98 (0.96, 0.99)	0.94 (0.89, 0.96)	0.95 (0.92, 0.97)
AI vs radiologist 1	0.98 (0.97, 0.99)	0.97 (0.94, 0.98)	0.97 (0.95, 0.98)	0.98 (0.96, 0.99)	0.97 (0.95, 0.98)	0.96 (0.92, 0.97)
AI vs radiologist 2	0.97 (0.96, 0.99)	0.98 (0.97, 0.99)	0.95 (0.92, 0.99)	0.97 (0.95, 0.98)	0.97 (0.94, 0.98)	0.98 (0.96, 0.99)
AI vs reference size (average of 2 radiologists)	0.98 (0.97, 0.99)	0.99 (0.98, 0.99)	0.97 (0.95, 0.98)	0.98 (0.96, 0.99)	0.99 (0.97, 0.99)	0.98 (0.96, 0.99)

Note that the artificial intelligence measurements incorporate a linear regression calibration toward the radiologists' mean values.
L-ACPA, left lung apex-to-costophrenic angle; L-AMD, left apex-to-diaphragm; pTLC, predicted total lung capacity; R-APCA, right lung apex-to-costophrenic angle; R-AMD, right apex-to-diaphragm.

verify that the measurements provided are accurate and appropriate for the image. However, we are currently working to develop an enhanced 3-step framework to replace the 2-step model presented in this study, which will not only report lung measurements but also identify abnormalities or technical limitations that may compromise measurement accuracy and thus warrant additional imaging to ensure more precise measurement determination.

Finally, although not the focus of this study, the use of pTLC as an alternative to direct measurement for donor/recipient size matching must also be addressed. Although pTLC is easily calculated, it not uncommonly will overestimate or underestimate volumes in recipients because it does not account for underlying disease, race, or age and has not been validated in different patient groups.[3,5,8,17] Proponents have argued that actual TLC measurements should be used to overcome this limitation; however, not all programs routinely collect this information—either as standard practice or because of recipient illness levels.[5,6] Although CXRs measurements have been shown to correlate well with actual TLC,[5,10,18] they are hindered by the aforementioned limitations. Although CT volumetry can be used to overcome some of these limitations, it is not usually obtained from transplant donors. CT measurements are also affected by mechanical ventilation settings, can be challenging to obtain due to donor instability, and require staff to perform who may not be available at off hours—likely limiting its adoption.

In summary, we have developed an automated system capable of consistently and efficiently delivering precise, standardized lung measurements from the portable chest radiographs universally used for lung transplant donor evaluation. The user-friendly nature and inherent scalability of our automated framework makes it a particularly valuable tool—capable of deployment even by resource-constrained remote hospitals without compromising accuracy. In future collaborative work with our organ procurement organization, we intend to explore having the onsite transplant coordinator deploy this system at the donor hospital. An additional advantage of AI-automated lung measurement is its ability to facilitate comparison between strategies for donor/recipient size matching—enabling the analysis of large imaging data sets collected as part of donor offers to better understand the strengths and weaknesses of each strategy. Future investigations should also prioritize addressing the limitations identified above to further improve the robustness and generalizability of our findings.

Author contributions

E.C., M.K.I., D.S., and R.R. conceived and designed the study. E.C., M.K.I., T.A., W.B.G., Y.S., A.R., A.S., S.Y., A.M., and A.A. were responsible for acquisition of data. M.K.I., E.C., D.S., and R.R. analyzed and interpreted the data. All authors drafted or revised the manuscript for important intellectual content and approved the final version to be published.

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Declaration of competing interest

The authors of this manuscript have conflicts of interest to disclose as described by the American Journal of Transplantation. W.B. Gefter receives grant support and consultant fees from Siemens Healthineers. E. Cantu receives grant support from XVIVO and consultant fees from United Therapeutics and CSL Behring. All other authors have no disclosures to report.

Data availability

Owing to the sensitive nature of the DICOMs used and the potential for loss of confidentiality, raw data will remain confidential and will not be shared. Code used for analysis has been made publicly available at https://pypi.org/project/AILungMeasure/.

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